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Data Structure and Algorithms

Final Project

Project Title :

< Flight Management System >

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2. Abstract / Summary :

The Flight Management System is a comprehensive solution for managing passenger information within the airline industry. It maintains records including passenger ID, passenger type (Business class, first class, or economy class).Passengers can purchase tickets by providing basic information such as name, passport number, ticket type, and payment details. The system supports ticket upgrades or downgrades post-purchase and allows passengers to print their information. With the capacity to handle a large number of passengers, the system enables the flight manager to efficiently search for passengers based on either passenger number or name, facilitating streamlined management and retrieval of passenger data.

3. Introduction :

Flight Management System stands as a comprehensive and adaptive solution, catering to the multifaceted needs of the airline industry. With its capacity to handle large passenger volumes, support ticket modifications, and facilitate expedited data retrieval, the FMS emerges as an indispensable tool for modern airline operations.

4. Problem Statement :

Is a simple airline management system that allows passengers to purchase tickets, upgrade or degrade their ticket class, and display their information. Additionally, flight managers can search for passengers by ID or name and display a list of all passengers.

5. Methodology :

- **Data structures used for the solution and why?**

The data structure used in our C++ program is a linked list to manage passenger information. Specifically, we've implemented a singly linked list where each node of the list represents a passenger, and the Passenger struct holds the relevant information about each passenger.

Linked List :

The Passenger struct represents a node in the linked list. It contains information about a passenger (name, address, etc.) and a pointer (link) to the next passenger in the list.

The linked list is created and managed using pointers (head, tail, temp, cur). head points to the first node, tail points to the last node, temp is used for creating new nodes, and cur is a temporary pointer used for traversing the list.

Dynamic Memory Allocation :

new Passenger is used to dynamically allocate memory for a new passenger node. This allows you to create nodes as needed during runtime.

The use of a linked list is suitable for this scenario because it provides efficient insertion and deletion operations, which are common when managing a list of passengers. Additionally, dynamic memory allocation allows you to manage memory more efficiently, allocating memory only when needed.

• Algorithm :

The program uses a menu-driven approach to interact with the user, allowing them to choose between being a flight passenger or a flight manager.

Nested menu structures are employed for passenger and manager services.

Main Menu :

1-Flight Passenger: Accesses the passenger menu.

2-Flight Manager: Accesses the manager menu.

3-Exit: Exits the program.

Passenger Service Algorithm :

The PassengerService function handles the passenger-related operations based on the user's choice (Purchase a ticket, Upgrade/Degrade a ticket, Display passenger information).

Specific functions (PurchaseATicket, UpgradeOrDegradeTicket, DisplayPassengerDetails, Go back to the main menu) are called based on the user's choice.

Manager Service Algorithm :

The ManagerService function handles the manager-related operations based on the user's choice (Search passenger by number, Search passenger by name, Display all passengers).

Specific functions (SearchPassengerByNumber, SearchPassengerByName, displayAllPassengers, Go back to the main menu) are called based on the user's choice.

Passenger Purchase Ticket Algorithm :

The PurchaseATicket function allows a passenger to enter their details, including ID, name, passport number, phone number, address, ticket number, and ticket degree.

It validates the entered ID to ensure uniqueness, and based on user input, assigns a ticket degree to the passenger.

Display Passenger Details Algorithm :

The DisplayPassengerDetails function prompts the user to enter their ID and searches the linked list for the corresponding passenger.

If found, it displays the details of that passenger.

Search Passenger Algorithm :

Both SearchPassengerByNumber and SearchPassengerByName functions search for a passenger based on the entered ID or name, respectively.

They traverse the linked list and display the details if a match is found.

Upgrade/Degrade Ticket Algorithm :

The UpgradeOrDegradeTicket function allows a manager to upgrade or degrade a passenger's ticket degree.

It displays the current ticket degree and prompts the manager to choose a new degree.

Display All Passengers Algorithm :

The displayAllPassengers function traverses the linked list and displays details for all passengers.

ID_Validation :

Validates if a given Id already exists in the system

Overall, the algorithmic structure revolves around handling user input, manipulating the linked list based on user choices, and displaying relevant information. The use of functions helps modularize the code and makes it more readable and maintainable.

- **Executable Program Code:**

FlightManagmentSystem_FinalDraft.cpp

6. Results and discussion :

The first thing you will see is a menu that asks whether you are a flight passenger, a flight manager, or if you want to exit the program.

```
cout << "----------------------------" << endl ;
cout << "---         MENU         ---" << endl ;
cout << "----------------------------" << endl ;

cout << "Welcome to our airlines! are you a flight passenger or a flight manager?" << endl ;
cout << endl ;
cout << "1) Flight Passenger. \n2) Flight Manager. \n3) Exit." << endl ;
cout << endl ;
```

```
----------------------------
---         MENU         ---
----------------------------
Welcome to our airlines! are you a flight passenger or a flight manager?

1) Flight Passenger.
2) Flight Manager.
3) Exit.
```

Passenger Menu :

Purchase a ticket: Allows passengers to input their details and purchase a ticket, including selecting the ticket class.

Valid Input:

Enter a unique ID that is not already in use.

Provide a name, passport number, phone number, address, and ticket number.

Choose a ticket degree by entering 1, 2, or 3 based on your options.

```
switch(choice)
{
    case 1 :
        {
            do
            {

                cout << "----------------------------" << endl ;
                cout << "---  PASSENGER SERVICES  ---" << endl ;
                cout << "----------------------------" << endl ;
                cout << endl ;
                cout << "1) Purchase a ticket. \n2) Upgrade or Degrade a ticket! \n3) Print Passenger's information.
                         \n4) Go back to main menu " << endl ;
                cout << endl ;
                cout << "Enter your choice: " ;
                cin >> PassChoice ;


                switch(PassChoice)
                {
                    case 1 :
                    case 2 :
                    case 3 :
                        {
                            PassengerService(PassChoice) ;
                            flag = true ;
                            break ;
                        }

                    case 4 :
                        {
                            flag = true ;
                            break ;
                        }

                    default :
                        {
                            cout << endl ;
                            cout << "Wrong entry! Try again!" << endl ;
                        }
                }


            }while(flag = false);

            break ;
        }
```

```
else
{
    cout << "ID is valid!" << endl ;
    temp->ID = id ;
}

}while(match == true) ;

cout<<"Please Enter Your Name: " ;
cin.ignore() ;
getline(cin, temp->name) ;

cout<<"Please Enter Your Passport Number: " ;
cin>>temp->PassportNumber;

cout<<"Please Enter Your Phone Number: " ;
cin>>temp->PhoneNumber;

cout<<"Please Enter Your Adderss: ";
cin>>temp->Address;

cout<<"Please Enter Your Ticket Number: " ;
cin>>temp->TicketNumber;

cout<<"Choose Your Ticket degree: " << endl ;

cout << "----------------------------" << endl ;
cout << "---   PASSENGER Ticket   ---" << endl ;
cout << "----------------------------" << endl ;

cout << "1. High Class     \t2500 SAR." << endl ;
cout << "2. business Class \t1750 SAR. " << endl ;
cout << "3. Economy Class \t800 SAR." << endl ;

cout << endl ;
```

```
----------------------------
---  PASSENGER SERVICES  ---
----------------------------

1) Purchase a ticket.
2) Upgrade or Degrade a ticket!
3) Print Passenger's informatiom.
4) Go back to main menu

Enter your choice: 1

***************** Passenger Details *****************
Please Enter Your ID:
101
ID is valid!
Please Enter Your Name: mohammed ali
Please Enter Your Passport Number: 114261
Please Enter Your Phone Number: 0502510552
Please Enter Your Adderss: altaawun
Please Enter Your Ticket Number: 4001
Choose Your Ticket degree:
----------------------------
---   PASSENGER Ticket   ---
----------------------------
1. High Class           2500 SAR.
2. business Class       1750 SAR.
3. Economy Class        800 SAR.

Enter your choice: 2
----------------------------------------
Passenger Ticket Degree is Business Class
----------------------------------------

Ticket is purchased successfully!
```

```
***************** Passenger Details *****************
Please Enter Your ID:
301
ID is valid!
Please Enter Your Name: yousef adam
Please Enter Your Passport Number: 317861
Please Enter Your Phone Number: 0529954110
Please Enter Your Adderss: alnaseem
Please Enter Your Ticket Number: 5028
Choose Your Ticket degree:
----------------------------
---   PASSENGER Ticket   ---
----------------------------
1. High Class           2500 SAR.
2. business Class       1750 SAR.
3. Economy Class        800 SAR.

Enter your choice: 1
-------------------------------------
Passenger Ticket Degree is High Class
-------------------------------------

Ticket is purchased successfully!
```

```
***************** Passenger Details *****************
Please Enter Your ID:
201
ID is valid!
Please Enter Your Name: salman ahmad
Please Enter Your Passport Number: 129465
Please Enter Your Phone Number: 0557233168
Please Enter Your Adderss: alkalidiya
Please Enter Your Ticket Number: 7211
Choose Your Ticket degree:
----------------------------
---   PASSENGER Ticket   ---
----------------------------
1. High Class           2500 SAR.
2. business Class       1750 SAR.
3. Economy Class        800 SAR.

Enter your choice: 3
----------------------------------------
Passenger Ticket Degree is Economy Class
----------------------------------------

Ticket is purchased successfully!
```

Invalid ID Input:

Enter an ID that already exists. This will trigger the ID validation mechanism in your program.

Cancel Purchase:

Choose to cancel the purchase when prompted after entering a duplicate ID.

```
***************** Passenger Details *****************
Please Enter Your ID:
201
The inserted ID is already existed!
Would you like to try another ID or cancel your purchase?
1) Try another ID.
2) Cancel.
Your option: 2

Canceling purchase!
```

```
cout<<"***************** Passenger Details *****************" << endl;

do
{

cout<<"Please Enter Your ID: "<< endl;
cin >> id ;
match = ID_Validation(id) ;

if(match == true)
{
    cout << "The inserted ID is already existed!" << endl ;
    cout << "Would you like to try another ID or cancel your purchase?" << endl ;
    cout << "1) Try another ID. \n2) Cancel." << endl ;
    cout << "Your option: " ;
    cin >> option ;

    cout << endl ;

    if(option == 1)
    {
        match = true ;
    }

    else if(option == 2)
    {
        cout << "Canceling purchase!" << endl ;

        delete temp ;

        return ;
    }

    else
    {
        cout << "INVALID ENTRY! TRY AGAIN!" << endl ;
        match = true ;
    }
}
```

Print Passenger's Information: Displays details of a specific passenger based on their ID.

```
void DisplayPassengerDetails()
{

    cur = head ;

    int id ;
    cout << "Enter your ID number: " ;
    cin >> id ;

    while(cur != NULL)
    {
        if(id == cur->ID)
        {
            cout << endl ;

            cout << "------ DETAILS FOR PASSENGER [ " << cur->name << " ] ------" << endl;

           cout << "----------------------" << endl ;

            cout << "Name " << cur->name << endl ;

            cout << "ID: " << cur->ID << endl ;

            cout << "Passport Number: " << cur->PassportNumber << endl ;

            cout << "Phone Number: " << cur->PhoneNumber << endl ;

            cout << "Adderss: " << cur->Address << endl ;

            cout << "Ticket Number: " << cur->TicketNumber << endl ;

            cout << "Ticket degree: " << cur->PassengerTicketDegree << endl ;

            cout << "----------------------" << endl ;

            return ;
        }
```

```
1) Purchase a ticket.
2) Upgrade or Degrade a ticket!
3) Print Passenger's informatiom.
4) Go back to main menu

Enter your choice: 3
Enter your ID number: 201

------ DETAILS FOR PASSENGER [ salman ahmad ] ------
----------------------
Name salman ahmad
ID: 201
Passport Number: 129465
Phone Number: 557233168
Adderss: alkalidiya
Ticket Number: 7211
Ticket degree: Economy Class
```

Upgrade or Degrade a ticket: Allows passengers to change their ticket class.

```
void UpgradeOrDegradeTicket(Passenger* passenger)
{
    int choice;

    cout << "Current Ticket Degree: " << passenger->PassengerTicketDegree << endl;
    cout << "Choose New Ticket Degree: " << endl;
    cout << "1. High Class" << endl;
    cout << "2. Business Class" << endl;
    cout << "3. Economy Class" << endl;

    cout << "Enter your choice: ";
    cin >> choice;

    switch (choice)
    {
        case 1:
            passenger->PassengerTicketDegree = "High Class";
            break;

        case 2:
            passenger->PassengerTicketDegree = "Business Class";
            break;

        case 3:
            passenger->PassengerTicketDegree = "Economy Class";
            break;

        default:
            cout << "Invalid choice! Ticket remains unchanged." << endl;
    }

    cout << "Ticket successfully upgraded or degraded!" << endl;
}
```

```
Welcome to our airlines! are you a flight passenger or a flight manager?

1) Flight Passenger.
2) Flight Manager.
3) Exit.

Enter your choice: 1
---------------------------
---  PASSENGER SERVICES  ---
---------------------------

1) Purchase a ticket.
2) Upgrade or Degrade a ticket!
3) Print Passenger's informatiom.
4) Go back to main menu

Enter your choice: 2
Enter passenger ID for ticket upgrade or degrade: 201
Current Ticket Degree: Economy Class
Choose New Ticket Degree:
1. High Class
2. Business Class
3. Economy Class
Enter your choice: 1
Ticket successfully upgraded or degraded!
```

```
Enter your choice: 3
Enter your ID number: 201

------ DETAILS FOR PASSENGER [ salman ahmad ] ------
----------------------
Name salman ahmad
ID: 201
Passport Number: 129465
Phone Number: 557233168
Adderss: alkalidiya
Ticket Number: 7211
Ticket degree: High Class
----------------------
```

Go back to the main menu: Returns to the main menu.

```
Enter your choice: 4
----------------------------
---          MENU          ---
----------------------------
Welcome to our airlines! are you a flight passenger or a flight manager?

1) Flight Passenger.
2) Flight Manager.
3) Exit.
```

```
cout << "-------------------------" << endl ;
cout << "--  PASSENGER SERVICES  --" << endl ;
cout << "-------------------------" << endl ;
cout << endl ;
cout << "1) Purchase a ticket. \n2) Upgrade or Degrade a ticket! \n3) Print Passenger's information. \n4) Go back to main menu " << endl ;
cout << endl ;
cout << "Enter your choice: " ;
cin >> PassChoice ;


switch(PassChoice)
{
    case 1 :
    case 2 :
    case 3 :
        {
            PassengerService(PassChoice) ;
            flag = true ;
            break ;
        }

    case 4 :
        {
            flag = true ;
            break ;
        }

    default :
        {
            cout << endl ;
            cout << "Wrong entry! Try again!" << endl ;
        }
}
```

Flight Manager

The program prompts the Flight Manager to choose from different options, such as searching for a passenger by ID, searching for a passenger by name, displaying all passengers, or going back to the main menu.

```
case 2 :
{
    do
    {

        cout << "----------------------------" << endl ;
        cout << "---   MANAGER SERVICES   ---" << endl ;
        cout << "----------------------------" << endl ;
        cout << endl ;
        cout << "1) Search Passenger with ID. \n2) Search passenger with name. \n3) Display all passengers.
        \n4) Go back to main menu. " << endl ;
        cout << endl ;
        cout << "Enter your choice: " ;
        cin >> ManChoice ;

        if(ManChoice == 1 || ManChoice == 2 || ManChoice == 3)
        {
            ManagerService(ManChoice) ;
            flag = true ;
        }

        else if(ManChoice == 4)
        {
            flag = true ;
            cout << endl ;
        }

        else
        {
            cout << endl ;
            cout << "Wrong entry! Try again!" << endl ;
        }

    }while(flag = false);

    break ;
}
```

```
Welcome to our airlines! are you a flight passenger or a flight manager?

1) Flight Passenger.
2) Flight Manager.
3) Exit.

Enter your choice: 2
----------------------------
---   MANAGER SERVICES   ---
----------------------------

1) Search Passenger with ID.
2) Search passenger with name.
3) Display all passengers.
4) Go back to main menu.
```

Search Passenger with ID: Allows managers to search for a passenger by their ID and display their information.

```
void SearchPassengerByNumber(int passengerNumber)
{
    Passenger *current = head;

    while (current != NULL) {
        if (current->ID == passengerNumber) {
            // Passenger found, display details
            cout << "Passenger found! Details:" << endl;
            cout << "  Passenger Name: " << current->name << endl;
            cout << "  Passenger ID: " << current->ID << endl;
            cout << "  Passenger Passport Number: " << current->PassportNumber << endl;
            cout << "  Passenger Phone Number: " << current->PhoneNumber << endl;
            cout << "  Passenger Address: " << current->Address << endl;
            cout << "  Passenger Ticket Number: " << current->TicketNumber << endl;
            cout << "  Passenger Ticket Degree: " << current->PassengerTicketDegree << endl;

            return ; // Exit the function after finding the passenger
        }

        current = current->link ;
    }

    // Passenger not found
    cout << "Passenger with ID " << passengerNumber << " not found." << endl;
}
```

```
Enter your choice: 1
Enter the Passenger ID to search: 101
Passenger found! Details:
  Passenger Name: mohammed ali
  Passenger ID: 101
  Passenger Passport Number: 114261
  Passenger Phone Number: 502510552
  Passenger Address: altaawun
  Passenger Ticket Number: 4001
  Passenger Ticket Degree: Business Class
```

```
Enter your choice: 1
Enter the Passenger ID to search: 401
Passenger with ID 401 not found.
```

Search passenger with name: Allows managers to search for a passenger by their name and display their information.

```
void SearchPassengerByName(const string& name)
{

    Passenger* current = head;
    bool found = false;

    while (current != NULL)
    {
        if (current->name == name)
        {
            found = true;
            // Display passenger details or perform other actions
            cout << "Passenger found!" << endl;
            cout << "  Passenger Name: " << current->name << endl;
            cout << "  Passenger ID: " << current->ID << endl;
            cout << "  Passenger Passport Number: " << current->PassportNumber << endl;
            cout << "  Passenger Phone Number: " << current->PhoneNumber << endl;
            cout << "  Passenger Address: " << current->Address << endl;
            cout << "  Passenger Ticket Number: " << current->TicketNumber << endl;
            cout << "  Passenger Ticket Degree: " << current->PassengerTicketDegree << endl;

            break; // Break out of the loop since the passenger is found
        }

        current = current->link;
    }

    if (!found)
    {
        cout << "Passenger with name '" << name << "' not found." << endl;
    }
}
```

```
Enter your choice: 2
Enter passenger name to search: yousef adam
Passenger found!
  Passenger Name: yousef adam
  Passenger ID: 301
  Passenger Passport Number: 317861
  Passenger Phone Number: 529954110
  Passenger Address: alnaseem
  Passenger Ticket Number: 5028
  Passenger Ticket Degree: High Class
```

```
Enter your choice: 2
Enter passenger name to search: salman ali
Passenger with name 'salman ali' not found.
```

Display all passengers: Displays information for all passengers.

```
Enter your choice: 3
***************** All Passengers Details *****************
***************** Passenger [ 1 ] Details *****************
----------------------
Name mohammed ali
ID: 101
Passport Number: 114261
Phone Number: 502510552
Adderss: altaawun
Ticket Number: 4001
Ticket degree: Business Class
----------------------
***************** Passenger [ 2 ] Details *****************
----------------------
Name salman ahmad
ID: 201
Passport Number: 129465
Phone Number: 557233168
Adderss: alkalidiya
Ticket Number: 7211
Ticket degree: High Class
----------------------
***************** Passenger [ 3 ] Details *****************
----------------------
Name yousef adam
ID: 301
Passport Number: 317861
Phone Number: 529954110
Adderss: alnaseem
Ticket Number: 5028
Ticket degree: High Class
----------------------
```

```cpp
void displayAllPassengers()
{
    cur = head ;
    int count = 1 ;

    if(head == NULL)
    {
        cout << "There are no passengers !!" << endl ;
    }

    else
    {
        cout << "***************** All Passengers Details *****************" << endl;

        while(cur != NULL)
        {
         cout<<"***************** Passenger [ "<< count << " ] Details *****************" << endl ;

            cout << "----------------------" << endl ;

            cout << "Name " << cur->name << endl ;

            cout << "ID: " << cur->ID << endl ;

            cout << "Passport Number: " << cur->PassportNumber << endl ;

            cout << "Phone Number: " << cur->PhoneNumber << endl ;

            cout << "Adderss: " << cur->Address << endl ;

            cout << "Ticket Number: " << cur->TicketNumber << endl ;

            cout << "Ticket degree: " << cur->PassengerTicketDegree << endl ;

            cout << "----------------------" << endl ;

            count++ ;

            cur = cur->link ;
        }

        cout << endl ;
    }
}
```

Exit: Exits the program.

```
----------------------------
---          MENU          ---
----------------------------
Welcome to our airlines! are you a flight passenger or a flight manager?

1) Flight Passenger.
2) Flight Manager.
3) Exit.

Enter your choice: 4

Invalid entry! Try again!
```

```cpp
do
{
    flag = false ;

    cout << "----------------------------" << endl ;
    cout << "---          MENU          ---" << endl ;
    cout << "----------------------------" << endl ;

    cout << "Welcome to our airlines! are you a flight passenger or a flight manager?" << endl ;
    cout << endl ;
    cout << "1) Flight Passenger. \n2) Flight Manager. \n3) Exit." << endl ;
    cout << endl ;
    cout << "Enter your choice: " ;
    cin >> choice ;
```

```
----------------------------
---          MENU          ---
----------------------------
Welcome to our airlines! are you a flight passenger or a flight manager?

1) Flight Passenger.
2) Flight Manager.
3) Exit.

Enter your choice: 3

Exiting the menu! Thank you for choosing our airlines!

------------------------------
```

```cpp
    case 3:
    {
        cout << endl ;
        cout << "Exiting the menu! Thank you for choosing our airlines!" << endl ;

        break ;
    }

    default :
        {
            cout << endl ;
            cout << "Invalid entry! Try again!" << endl ;
        }
}
```

7. Conclusion :

All in All, the development of this Airline Management System marks a significant stride in improving the efficiency and organization of airline operations. The program employs a linked list data structure to adeptly manage passenger details, fostering smooth interactions between passengers and managers. Through a menu-driven approach, the system accommodates a range of functionalities for both passengers and managers, including ticket purchasing, upgrades or downgrades, and comprehensive searches. The modular and readable code structure, coupled with effective error handling, contributes to a user-friendly experience. As the program leverages dynamic memory allocation and tailored algorithms, it exemplifies the potential of innovative solutions in simplifying and enhancing the management of airline services.

8. References :

Data Structures and Algorithms in C++